DHIRAJ ATHREYA H

+91 9620800820 — dhiraj.athreyas@gmail.com — linkedin.com/in/dhiraj-athreyas — Bangalore, India

SUMMARY

AI/ML Engineer with 1+ year of production experience building and deploying Computer Vision, Speech AI, and Generative AI systems for the Indian Navy. Proven track record of architecting GPU-accelerated multi-camera video analytics pipelines that scale to 40 concurrent RTSP streams while delivering 8–12× latency reduction. Skilled in designing hybrid RAG systems, multilingual speech pipelines, and LLM-powered applications using NVIDIA Triton, TensorRT, GStreamer, and modern open-source frameworks. Passionate about turning cutting-edge research into reliable, high-performance production systems.

TECHNICAL SKILLS

Programming: Python, SQL
Computer Vision & Deep Learning: YOLO, RF-DETR, InsightFace, ResNet50, OpenCV, MediaPipe, PyTorch, TensorFlow
NVIDIA & Inference: NVIDIA DeepStream 8.0, GStreamer, TensorRT, NVIDIA Dynamo Triton Inference Server
Generative AI, Agentic AI & NLP: RAG, LangChain, LangGraph, AI Agents, LLM Evaluation, Hugging Face Transformers, BM25, Semantic Search, Cross-Encoder Reranking, Speaker Diarization, Neo4j
Speech & Multilingual AI: Faster-Whisper, Silero VAD, SeamlessM4T, NLLB, M2M100, VITS
Databases & Vector Search: PostgreSQL, pgvector, MilvusDB, Qdrant, ChromaDB, MongoDB, Oracle SQL
Deployment & Tools: Docker, Nginx, Git, NumPy, Pandas, Scikit-learn, Matplotlib, Power BI

PROFESSIONAL EXPERIENCE

AI/ML Engineer Jan 2026 – Present
BluParrot — Indian Navy Incubation Centre for AI (Client) — Bangalore, India

- Architected and scaled a real-time multi-camera Vision AI pipeline for person detection, recognition, and tracking, evolving from CPU/OpenCV decoding to GPU-accelerated GStreamer, multiprocessing, and Triton Inference Server; validated scalability across **40 concurrent RTSP streams** while maintaining **¡20 GB VRAM** utilization.
- Optimized the end-to-end video analytics architecture by offloading H.265 decoding to the GPU, centralizing **RF-DETR** and InsightFace inference with Triton dynamic batching, and integrating Qdrant HNSW-based face retrieval; reduced measured latency from **800–3000+ ms** to **100–250 ms** at **15 concurrent streams**, while improving pipeline scalability and stream-level fault isolation.
- Developed multilingual speech-processing pipelines for language identification, transcription, translation, and speech synthesis using Faster-Whisper, NLLB, SeamlessM4T, VITS, and Silero VAD; improved language identification accuracy to **82.35%** (∼15% improvement) on noisy maritime audio, reduced direct speech-to-English translation latency by **54%** (processing 135 s of audio in 5.14 s, **26.3× real-time**), and optimized cascaded translation throughput to **32 sentences/s** with **5.5×** lower memory using FP16 quantization.
- Built a Generative AI Meeting Assistant for speaker diarization, transcription, summarization, and structured minutes-of-meeting generation; evaluated open-source LLMs across multiple parameter scales for accuracy and contextual coherence.
- Engineered a multi-format RAG pipeline for uploaded documents and web-scraped content, implementing **BM25 + dense semantic retrieval + cross-encoder reranking** with PostgreSQL/pgvector vector storage and Neo4j knowledge graphs for structured semantic retrieval.
- Containerized and deployed the complete AI application stack using Docker and Nginx, resolving deployment and integration issues to ensure reliable end-to-end operation.

> **Represented the INDIAN NAVY as an Exhibitor and Delegate at the India AI Impact Expo 2026, New Delhi**

AI Research Engineer Aug 2025 – Dec 2025
Indian Navy Incubation Centre for Artificial Intelligence — Bangalore, India (Onsite)

- Evaluated state-of-the-art computer vision approaches from **CVF** and **IEEE** literature for multi-camera surveillance, person re-identification, multi-object tracking, camera calibration, and geometric localization, establishing the research foundation for the project's production AI pipeline.
- Designed and implemented the initial multi-camera perception stack by benchmarking and integrating **YOLO**, InsightFace, TensorRT, **ResNet50**, and unified 2D mapping, establishing the foundation for the production-grade surveillance platform.
- Researched and developed camera self-calibration, feature extraction, and appearance-based person re-identification techniques, culminating in a custom **ResNet50 ReID** pipeline for robust cross-camera identity association.

- Co-authored a research paper documenting the system architecture, experimental methodology, and technical findings, translating research outcomes into an engineering blueprint for subsequent production deployment.

PROJECTS

Kautilya — OSINT Scraping and Ingestion with RAG Application Current

- Built an end-to-end multilingual OSINT pipeline for web acquisition, ingestion, vector storage and RAG-based intelligence retrieval, combining BM25 keyword matching with semantic search.

Vachaspathi — Multilingual AI Tutor Current

- Developed a multilingual AI educational assistant using LLM-generated adaptive explanations, VITS text-to-speech and M2M100/NLLB translation.

Sports Injury Recovery Period Prediction Nov 2023 – Apr 2024

- Built a Flask-based ML application for predicting athlete injury recovery duration; performed feature engineering, preprocessing and model tuning, and designed the supporting MySQL schema.

EDUCATION

M.Tech in Data Science and AI 2025 – 2027

Indian Institute of Information Technology (IIIT), Dharwad • Current status: 3rd semester • GPA: 8.35

B.E. in Computer Science and Engineering Dec 2020 – May 2024

Dayananda Sagar Academy of Technology and Management — Bangalore, India • GPA: 8.76/10

ACADEMIC ACHIEVEMENT

- **Fantom Code Hackathon** — Team Leader, 4-member team — National-level 24-hour hackathon, RV Institute of Technology and Management, Bengaluru.